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Posted Date: June 13th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-6699075/v1

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Additional Declarations: There is **NO** Competing Interest.

AI-Powered Standard Operating Procedure Generation and Optimization Using Large Language Models and Chroma Databases in Chemistry

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Abstract

Creating clear and accurate standard operating procedures is essential for safety and consistency in chemistry laboratories and industrial settings. These documents guide users through equipment handling, experimental steps, and safety protocols, but are often time-consuming to produce and update. In this article, we describe a system that uses recent advances in artificial intelligence to assist in generating these procedures. The system retrieves relevant information from a curated database of chemical manuals, regulatory guidelines, and scientific texts, and uses it to generate customized documents based on user input. Users can select which sections to include, such as calibration methods or waste disposal steps, ensuring the output aligns with specific needs. Designed to minimize errors common in automated text generation, the system helps maintain accuracy and relevance. This approach simplifies documentation tasks in chemistry while supporting compliance and safety, offering a practical tool for researchers and professionals in laboratory.

Keywords: Standard Operating Procedure (SOP), Generative AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Chroma Database, AI-driven, Laboratory Safety, Regulatory Compliance, Chemistry Workflows

1 Introduction

In chemistry, SOPs serve as the foundation for maintaining experimental consistency, laboratory safety, and regulatory compliance. Chemists rely on well-defined SOPs to standardize procedures, ensuring reproducibility in research and industrial applications. Precise documentation of chemical handling, synthesis methods, and analytical techniques is essential not only for accuracy but also for adherence to safety protocols, particularly when working with hazardous substances. Regulatory bodies such as the Occupational Safety and Health Administration (OSHA) and the Environmental Protection Agency (EPA) mandate strict adherence to SOPs to mitigate risks associated with chemical exposure, waste disposal, and process safety [10]. However, traditional SOP documentation remains a labor-intensive and error-prone process, often requiring significant manual effort to align with evolving regulatory frameworks and advancements in chemical methodologies. This challenge highlights the necessity for an automated, intelligent system capable of dynamically generating and optimizing SOPs to support chemists in maintaining high standards of laboratory operation.

The integration of AI-powered language models into SOP generation presents a transformative approach to addressing these challenges. By leveraging retrieval-augmented generation (RAG) and Chroma databases, AI-driven SOPs can dynamically pull information from established chemical literature, equipment manuals, and safety guidelines, ensuring that generated procedures remain scientifically sound and up-to-date. This capability is particularly valuable in fields such as analytical chemistry, where methods must adhere to Good Laboratory Practice (GLP) guidelines, or in synthetic chemistry, where reaction conditions must be meticulously documented to ensure reproducibility and safety [11]. Furthermore, AI-driven SOP optimization allows chemists to customize procedural elements, incorporating real-time hazard assessments and regulatory updates, thereby reducing the risk of non-compliance and enhancing overall laboratory efficiency. By automating SOP creation while preserving the accuracy and reliability demanded in the chemical sciences, this approach establishes a new paradigm for laboratory documentation, bridging the gap between traditional chemical methodologies and modern AI advancements.

The loss of an SOP in a laboratory setting can lead to significant disruptions, impacting research continuity, safety, and compliance. Common scenarios where SOPs may be misplaced include accidental deletion from digital records, damage to physical copies due to chemical spills, or personnel turnover where critical procedural knowledge is not effectively transferred. In regulated environments such as pharmaceutical and analytical chemistry laboratories, the absence of an SOP can halt experimental workflows, delay production, and even lead to regulatory violations [17].
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workflows, delay production, and even lead to regulatory violations [17]. Additionally, unexpected equipment failures may require immediate reference to maintenance or calibration SOPs, which, if unavailable, could extend downtime and compromise data integrity [8]. To mitigate these risks, an AI-powered SOP generation system becomes essential, offering instant retrieval and customization of procedural documents. By leveraging pre-existing chemical databases and regulatory guidelines, such a system can regenerate lost SOPs with high accuracy, ensuring that laboratory operations remain uninterrupted and compliant with safety and quality standards. This approach not only enhances operational resilience but also fosters a more adaptive and efficient laboratory environment, reducing dependency on static documentation and enabling real-time procedural adjustments when necessary.

The remainder of this article is structured as follows: Section 2 provides a comprehensive review of existing work on SOP generation and AI-driven documentation. Section 3 presents the methodology, detailing data collection, model integration, embedding strategies, and the implementation of RAG framework. Section 4 outlines the validation and quality control measures applied to assess the system's accuracy and reliability. Section 5 introduces the user interface developed for enhanced SOP generation, providing an intuitive experience for users. Finally, Section 6 concludes the paper by summarizing key findings, discussing the limitations, and proposing directions for future research.

2 State of Art

2.1 Standard Operating Procedures in Chemistry

Standard Operating Procedures (SOPs) play a fundamental role in ensuring experimental reproducibility, regulatory compliance, and safety in chemical research and industrial applications. Properly structured SOPs help in minimizing variability in chemical processes, ensuring adherence to international guidelines such as ISO 9001 and Good Laboratory Practices (GLP), and reducing hazards associated with handling chemicals and equipment [5]. However, manual SOP creation is often time-consuming, prone to inconsistencies, and difficult to maintain as procedures evolve with new scientific discoveries and technological advancements [9].
In many laboratory settings, outdated SOPs can lead to misuse of chemicals, improper equipment calibration, and non-compliance with safety standards [14]. Additionally, lost or inaccessible SOPs can disrupt laboratory workflows, delaying critical experiments and increasing the risk of accidents [16]. These challenges highlight the need for automated and dynamic SOP management systems capable of adapting to evolving laboratory needs.

2.2 AI-Driven Documentation and Automated SOP Generation

Recent advances in Artificial Intelligence (AI), particularly in Natural Language Processing (NLP) and Generative AI (GenAI), have paved the way for automating complex documentation tasks, including SOP generation. Large Language Models

(LLMs) such as GPT-4, BERT, and T5 have demonstrated the ability to generate technical and regulatory documents with high accuracy and contextual relevance [3], [12]. However, one of the primary concerns in using AI-generated documentation in scientific fields is the risk of hallucinations, where AI models produce factually incorrect or misleading information [6]. To enhance the reliability and domain-specific accuracy of AI-generated SOPs, the Retrieval-Augmented Generation (RAG) framework has emerged as a powerful approach [7]. By integrating a Chroma database, AI models can retrieve validated scientific literature, regulatory documents, and equipment manuals, ensuring that SOPs are generated based on authentic and up-to-date sources. This hybrid approach not only improves content precision but also enables customizable SOP generation, allowing users to specify required sections such as safety protocols, calibration procedures, waste disposal guidelines, and experimental steps [2].

2.3 The Need for AI-Powered SOP Solutions in Chemistry

Despite the advancements in AI-driven text generation, domain-specific SOP automation in chemistry remains an underexplored area. Existing document management systems primarily focus on SOP storage and retrieval rather than automated generation [4]. Some AI-assisted documentation tools are available in the healthcare and pharmaceutical industries [15], but they lack the flexibility required for chemistry-specific workflows, where experimental conditions, materials, and safety measures vary significantly. This paper introduces an AI-powered SOP generation platform tailored specifically for chemistry laboratories. By leveraging LLMs, RAG, and Chroma databases, the proposed system ensures that SOPs remain accurate, up-to-date, and customizable. This approach not only streamlines documentation workflows but also enhances laboratory safety, regulatory compliance, and operational efficiency.

3 Methodology

The methodology outlined in this work represents a systematic approach to leveraging advanced AI tools and frameworks for the generation of SOPs. Figure 1 provides an overview of the workflow, illustrating the integration of data collection, processing, semantic embedding, and retrieval-augmented generation (RAG) techniques to produce accurate and contextually relevant SOPs. Each stage of the process is designed to ensure efficiency, scalability, and high-quality outputs, addressing the challenges associated with SOP generation.

3.1 Data Collection

One of the most challenging aspects of this work was gathering data for Standard Operating Procedures (SOPs) from a variety of sources. SOPs, often referred to as the blueprint or instruction manual for how tasks are executed in an organization, are critical documents that outline step-by-step processes to ensure consistency, efficiency, and compliance. They serve as a reference for employees, contractors, and stakeholders to follow established procedures accurately. However, despite their importance, collecting high-quality and diverse SOP data is notoriously difficult due to issues such as

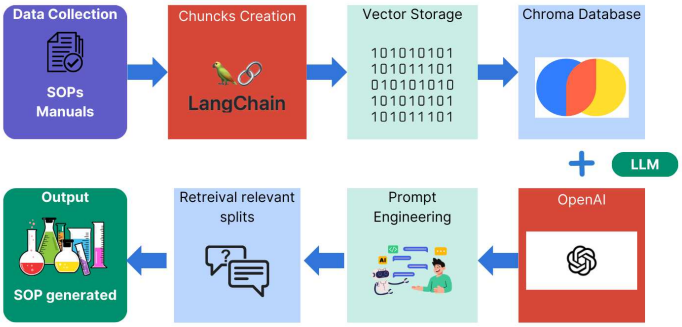

Figure 1: Methodology adapted for SOP generation

access restrictions, variability in formats, and the ever-changing nature of operational requirements.

In this work, the SOPs were collected using multiple methods to build a comprehensive database suitable for automated SOP generation. Online resources served as a primary source whenever available, providing access to a range of publicly shared SOPs. Additionally, significant contributions were made through collaboration with colleagues and industry professionals who work directly with these procedures. Requests for SOPs from trusted connections added valuable diversity and authenticity to the database by including real-world examples from various organizations and industries.

Recognizing the limitations of relying solely on traditional SOP documents, this work also incorporated equipment manuals as a supplemental data source. Equipment manuals, which are widely available and often more detailed, provide step-by-step instructions for operating and maintaining specific devices. These manuals not only align closely with the structure and intent of SOPs but also offer technical depth and accuracy. By including these manuals alongside SOPs, the database was enriched with reliable, contextually relevant information, addressing one of the critical flaws in traditional SOP systems: the risk of hallucination. In the context of Generative AI, hallucination refers to the generation of plausible-sounding but factually incorrect or unsupported content. The inclusion of equipment manuals mitigates this risk by providing a robust factual grounding for the model during the retrieval and generation phases.

Moreover, the integration of equipment manuals allowed for the enhancement and expansion of the database, filling gaps where SOPs were unavailable or difficult to obtain. Manuals are typically written by experts with a deep understanding of the equipment and processes involved, ensuring high-quality data. Their standardized format and detailed descriptions make them an ideal supplement for SOP data, especially for use cases requiring high levels of technical accuracy and specificity.

3.2 Content Division and Model Integration into LangChain

The process begins with gathering and loading all relevant documents required for building the SOP generation system. This step is executed using the load_documents function, which utilizes the PyPDFDirectoryLoader from LangChain. LangChain is a powerful framework designed to assist in creating applications that leverage large language models (LLMs) [1]. It simplifies the integration of various data sources, processing pipelines, and retrieval mechanisms into a unified workflow. Specifically, LangChain's PyPDFDirectoryLoader is used to scan a specified directory (DATA_PATH) and extract the content of all PDF files contained within it.
The extracted content is then converted into Document objects, a standardized format provided by LangChain that includes metadata such as file names and page numbers along with the text content. This standardization ensures consistent handling of diverse file formats and structures, which is essential for downstream processing. Moreover, LangChain's modular design allows seamless integration with other components, such as text splitters, vector databases, and LLMs, making it an ideal choice for complex workflows like SOP generation. By leveraging LangChain's capabilities in this step, the system efficiently builds a robust dataset that serves as the foundation for the SOP generation pipeline. This framework not only streamlines the loading process but also provides a scalable solution for managing large volumes of heterogeneous documents, ensuring that the data is well-organized and ready for subsequent operations such as chunking, embedding, and semantic retrieval. This step is vital for creating a seamless pipeline where data can flow consistently and accurately across all stages of the process.

3.3 Embedding Documents into a Vector Space

Once the SOP documents are divided into manageable chunks, the next step involves embedding these chunks into a high-dimensional vector space. This is achieved using OpenAI's advanced embedding functions, which transform the textual data into numerical vectors. These vectors are designed to capture the semantic meaning of the content, allowing for an in-depth understanding of the relationships between different chunks. The embedding process is a critical component of the system, as it provides the foundation for semantic retrieval. By representing text as vectors in a high-dimensional space, the system enables efficient comparison and ranking of chunks based on their relevance to user queries. This transformation not only preserves the contextual meaning of the original text but also enhances the system's ability to retrieve information accurately and quickly.

3.4 Vector Store Integration with Chroma

After embedding the document chunks into a high-dimensional vector space, the next step involves storing and managing these embeddings in Chroma, a highperformance vector database [13]. Chroma plays a pivotal role in the system by enabling efficient semantic retrieval of relevant information. Using the add_to_chroma function, the embeddings are stored in a structured and searchable format, alongside their metadata, which includes file names, page numbers, and contextual tags. This metadata

ensures that each retrieved chunk remains traceable to its original source, enhancing the reliability and transparency of the system.

When a user query is issued, it is first transformed into an embedding using the same OpenAI embedding function employed during the document embedding phase. This ensures that both the query and the stored embeddings exist within the same semantic space. Chroma then performs similarity searches, comparing the query embedding to the stored vectors using metrics such as cosine similarity. The results are ranked based on relevance, ensuring that the most pertinent chunks are retrieved and passed to the subsequent generation phase.

Chroma's capabilities extend beyond retrieval efficiency. Its scalable architecture allows it to handle large volumes of data, making it suitable for managing the growing complexity of SOP databases. Furthermore, Chroma supports dynamic updates, enabling new embeddings to be added or outdated ones to be removed seamlessly. This adaptability ensures that the system remains current and responsive to evolving datasets while generating SOP.

3.5 Building the RAG System

The Retrieval-Augmented Generation (RAG) framework forms the backbone of the SOP generation system, seamlessly combining semantic retrieval and generative capabilities. After the document chunks are embedded and stored in the Chroma vector database, the RAG system uses semantic similarity metrics to identify and retrieve the most contextually relevant chunks based on user queries. These chunks are then augmented with supplemental information extracted from equipment manuals and other trusted sources to ensure the input to the generative model is comprehensive and reliable. The RAG system relies on OpenAI's API to power its language model, leveraging its advanced generative capabilities to transform the retrieved and augmented information into detailed SOP sections. By dynamically integrating retrieval and generation, the RAG system ensures that the generated SOPs are accurate, user-focused, and tailored to the specific needs of chemical laboratory workflows.

3.6 Prompt engineering

Prompt engineering is a pivotal step in the SOP generation pipeline, enabling the large language model (LLM) to transform retrieved content into coherent, contextually accurate, and highly detailed SOP sections. Following the retrieval of semantically relevant chunks from the Chroma vector database, carefully designed prompts are used to guide the LLM in structuring the output according to predefined SOP requirements. These prompts are tailored specifically for the chemical domain, incorporating detailed section templates such as Scope, Background, Safety, Calibration, and Quality Control, ensuring that all critical aspects of chemical laboratory workflows are addressed.

Each prompt is meticulously designed to include contextual instructions derived from the retrieved chunks and any additional information extracted from equipment manuals. For example, when generating a calibration section, the prompt specifies exact procedural details, avoiding vague instructions like "refer to the manufacturer's manual." Instead, it requests step-by-step actions, including specific buttons to

press on the instrument interface, ensuring the generated SOP is both actionable and domain-specific. This grounding in factual data significantly reduces the risk of hallucinations—a common issue in generative AI—by ensuring the model's output remains firmly tied to the retrieved context.

The prompts are not static; they are dynamically structured to provide flexibility for customization. Users can specify whether to include or exclude certain sections, such as waste disposal protocols or safety guidelines, tailoring the generated SOP to their specific requirements. Additionally, the prompts are optimized to enforce consistency and clarity, ensuring the outputs maintain a professional tone while aligning with regulatory standards. This is particularly crucial in the chemical industry, where precise instructions and compliance are non-negotiable.

To enhance usability, the prompts adopt a modular and iterative approach. Each SOP section is generated independently, allowing for seamless integration into the final document. Prompts are designed to elicit responses that include technical depth, examples, and practical applications, ensuring that even complex procedures are explained in an accessible and step-by-step manner. For instance, safety sections may detail specific risks associated with chemicals and equipment, supplemented by clear instructions for handling hazardous materials.

4 Validation and Quality Control

To validate the reliability and accuracy of our SOP generation system, a controlled scenario was designed and executed. This scenario involved an existing SOP, for which all details and information were fully known and documented. The SOP was then deliberately hidden from the system, and all references, data, and related information about it were removed from the database and document repository. This ensured that the system had no direct knowledge or prior exposure to the target SOP. The objective was to evaluate whether the system, following the steps outlined in the methodology, could regenerate a comparable SOP purely through its retrieval, augmentation, and generation pipeline.

After removing the SOP, the program was run through all steps of the methodology. It loaded and split documents, generated embeddings, stored them in Chroma for semantic retrieval, and constructed the SOP using the RAG framework and prompt engineering. The generated SOP was then compared to the original hidden SOP to evaluate how accurately the system recreated the document. This comparison was achieved using a similarity calculation function based on TF-IDF (Term Frequency-Inverse Document Frequency) vectorization and cosine similarity, implemented as follows:

1. Extracting Text Content: The text content of both the original SOP and the system-generated SOP was extracted. Depending on the file format (either .docx or .pdf), specific extraction methods were applied to ensure consistent text representation.
2. Text Vectorization: TF-IDF vectorization was applied to transform the textual data into numerical vectors. This technique assigns weights to terms in the document, with higher weights given to terms that are more important (i.e., terms that appear

frequently in one document but rarely across others). This representation captures the unique characteristics of each document.

3. Cosine Similarity Calculation: Once the documents were vectorized, cosine similarity was used to measure the degree of similarity between the two vector representations. Cosine similarity calculates the cosine of the angle between two vectors in a high-dimensional space, with values ranging from -1 to 1:

   - A value of 1 indicates perfect similarity (identical vectors).
   - A value of 0 indicates no similarity (orthogonal vectors).
   - A value of -1 indicates complete dissimilarity.

   For this system, cosine similarity focused on positive correlations, meaning that higher values (closer to 1) indicated greater similarity between the original and generated SOPs.

4. Validation Results: The similarity scores for different SOPs across various materials and procedures consistently ranged between 0.8 and 0.9 in this work, demonstrating a high degree of alignment between the generated and original SOPs. This range indicates that the system effectively reconstructed the critical content, structure, and procedural details of the original SOPs, despite having no direct information about them.

5 User Interface for Enhanced SOP Generation

The Enhanced SOP Generator interface is a user-friendly tool designed to facilitate the creation of detailed and professional SOPs without requiring programming expertise. Built using the Python Tkinter library, the GUI provides a seamless experience for users, allowing them to generate customized SOPs for various instruments and procedures.

The interface includes the following key components:

- Instrument-Specific Inputs: Users can specify the instrument's name, brand, and model. These details ensure that the generated SOP is tailored to the equipment being used, enhancing its relevance and accuracy.
- Manual Upload Feature: Users can upload optional manuals in PDF format, which are processed and integrated into the SOP generation pipeline. This feature enriches the contextual database, ensuring the SOPs include highly relevant and detailed content from external references.
- Material Type Selection: A dropdown menu allows users to select the material type (e.g., liquid or solid). This input is incorporated into the SOP to refine procedural details and ensure compatibility with the specified material properties.
- Section Selection: Users can select which sections of the SOP to include, such as "Scope," "Safety," "Materials and Equipment," "Calibration," and "Procedure." A "Select All" option is also available for comprehensive SOP generation. The modularity ensures flexibility in creating SOPs specific to the user's needs.
- Output Path Specification: Users can choose where to save the final SOP document in Word format. This functionality ensures that the output is easy to access and share.

- Professional Customization: The system supports the inclusion of a company logo, adding a professional touch to the generated SOPs. The logo is dynamically included in the Word document header, enhancing branding.
- Integration with Advanced AI Models: The backend utilizes LangChain for document loading and splitting, Chroma for semantic search and retrieval, and OpenAI models for SOP content generation. These technologies ensure the generated SOPs are precise, contextually relevant, and adhere to professional standards.
- Error Handling and Guidance: The interface includes validation for required fields and provides clear error messages if essential information is missing. This feature makes the tool accessible even for non-technical users.

The SOP Generator empowers users to streamline the creation of SOPs, saving time and ensuring compliance with industry standards. With its intuitive design, dynamic features, and robust AI-driven backend, this tool bridges the gap between technical documentation requirements and practical

6 Conclusion and Future Work

In conclusion, this work presents a comprehensive and innovative approach to automating Standard Operating Procedure (SOP) generation using advanced AI frameworks. By integrating data collection from diverse sources, semantic embedding, and retrieval-augmented generation (RAG) techniques, the proposed system is capable of producing accurate and contextually relevant SOPs tailored to the rigorous demands of chemical laboratory workflows. The system leverages state-of-the-art tools such as LangChain for document processing, OpenAI's embedding functions for semantic representation, and the Chroma vector database for efficient retrieval, thereby ensuring that generated SOPs are both reliable and up-to-date. The incorporation of equipment manuals as supplementary data sources further enriches the dataset, significantly mitigating the risk of hallucinations—a common challenge in generative AI applications.

Validation of the system using a controlled scenario demonstrated that the regenerated SOPs achieved similarity scores ranging from 0.8 to 0.9 when compared to the original documents, coming from the manufacturing companies. These results underscore the effectiveness of the RAG framework in reconstructing critical content, structure, and procedural details, despite the absence of direct input data about the hidden SOP. Moreover, the modular design of the pipeline—from data collection and content division to prompt engineering—ensures not only high-quality outputs but also scalability and adaptability to evolving laboratory requirements. The integration of a user-friendly interface, built with the Python Tkinter library, further enables non-technical users to generate customized SOPs seamlessly, thereby bridging the gap between complex technical documentation and practical laboratory needs.

Despite these promising outcomes, several areas remain for further enhancement and exploration. One of the primary challenges is ensuring the continual accuracy and relevance of generated SOPs in a domain characterized by rapid technological and

regulatory changes. While the current system effectively minimizes hallucination risks through rigorous data grounding, future iterations could benefit from incorporating real-time data feeds and dynamic updates to reflect the latest regulatory standards and scientific discoveries.

Future work will focus on several key directions:

1. Enhanced Data Integration and Real-Time Updates: Future research should explore the integration of real-time data sources, such as online regulatory updates, IoT sensor data, and continuously updated scientific databases. This would enable the system to generate SOPs that reflect the most current practices and standards, thereby further reducing the risk of outdated procedures and ensuring compliance with evolving regulations.

2. Improving Domain Adaptation and Customization: Although the current system is tailored for chemical laboratory workflows, extending the framework to support multiple domains—such as pharmaceuticals, biotechnology, and industrial manufacturing—could significantly broaden its applicability. Future work may involve fine-tuning language models with domain-specific datasets and developing customizable templates that accommodate the unique requirements of various industries.

3. Advanced Prompt Engineering and Model Optimization: Continued refinement of prompt engineering strategies is essential to enhance the specificity and technical depth of generated SOP sections. Future iterations could incorporate adaptive prompt generation based on user feedback and domain expert evaluations. Additionally, exploring ensemble approaches that combine outputs from multiple models might further reduce the likelihood of hallucinations and improve overall output quality.

4. User-Centric Enhancements and Usability Studies: The development of the enhanced SOP generator interface marks a significant step toward democratizing the use of advanced AI in technical documentation. Future work should involve conducting extensive user studies and usability testing to gather feedback from laboratory personnel and domain experts. Such studies will be crucial in refining the interface, ensuring that it meets the practical needs of end-users, and identifying any bottlenecks or areas for improvement in the workflow.

5. Extended Validation and Benchmarking: While the current validation results are promising, additional benchmarking against industry-standard SOPs across a wider range of laboratory scenarios will be necessary. Future research should focus on establishing robust evaluation frameworks that incorporate both quantitative metrics (e.g., similarity scores, retrieval precision) and qualitative assessments (e.g., expert reviews) to comprehensively validate the system's performance.

6. Integration with Automated Compliance Monitoring Systems: Finally, integrating the SOP generation platform with automated compliance monitoring and reporting systems could offer substantial benefits. By ensuring that SOPs are not only generated but also continuously monitored for adherence to safety and regulatory standards, organizations can achieve higher levels of operational efficiency and risk management.

This work lays a solid foundation for the automated generation and optimization of SOPs using AI-powered language models and vector database integration. The promising results, coupled with the system's modular design and user-friendly interface, demonstrate the potential of this approach to revolutionize documentation practices in chemical laboratories. Future research will aim to build upon these advances, addressing current limitations and expanding the system's capabilities to meet the increasingly complex demands of modern laboratory environments.

Acknowledgments

This research was supported by Al Yamamah University, and King Fahd University of Petroleum & Minerals (KFUPM). The authors sincerely appreciate the continuous support and resources provided by both institutions, which have been instrumental in the successful completion of this work. We extend our gratitude to our colleagues and research teams for their valuable insights and constructive discussions.

Data Availability

The datasets, manuals, and code generated during the current study are publicly available on Zenodo at the following DOIs:

- **SOP Dataset**: Elias Mallouhy, R. (2025). *SOP Dataset* [Data set]. Zenodo. https://doi.org/10.5281/zenodo.15581784
- **Equipment Manuals**: Elias Mallouhy, R. (2025). *Manual-Dataset*. Zenodo. https://doi.org/10.5281/zenodo.15581822
- **Code (Jupyter Notebooks)**: Elias Mallouhy, R. (2025). *Code-Jupyter Notebook*. Zenodo. https://doi.org/10.5281/zenodo.15581829

These resources include sample Standard Operating Procedures (SOPs), equipment manuals, and the code used for the AI-powered SOP generation system, including LangChain and Chroma implementations. The repository structure allows other researchers to replicate, adapt, and extend the methods and findings reported in this study. **Correspondence and requests for additional materials** should be addressed to Dr. Roxane Elias Mallouhy (r_mallouhy@yu.edu.sa).

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